

Solar Sound System Simulator — macOS Build

This folder contains everything needed to produce a native macOS `.dmg` installer from the Python source.

What you need (on a Mac)

Requirement	Notes
macOS 11 Big Sur or newer	Older versions may work but are untested
Python 3.9 – 3.12 (system or Homebrew)	Must include Tkinter. Check: <code>python3 -c "import tkinter"</code>
py2app	Install once — see below
Xcode Command Line Tools	<code>xcode-select --install</code>

One-time setup

```
# 1. Install py2app into the project's venv (or system Python)
cd /path/to/Bike           # the repo root
python3 -m venv .venv
source .venv/bin/activate
pip install py2app
```

Tkinter on macOS

The macOS system Python **does not** include Tkinter.

Use the [python.org installer](https://python.org/installer) (includes Tk)

or Homebrew: `brew install python-tk@3.12`

Build the .dmg

```
cd ForMac
chmod +x build_dmg.sh
./build_dmg.sh
```

The script will:

1. Run `py2app` to bundle `app.py` + `rules_engine.py` + `simulate.py` into `ForMac/dist/Solar Sound System Simulator.app`
2. Create a drag-and-drop `.dmg` at `ForMac/dist/SolarSoundSimulator.dmg`

Install

1. Open `SolarSoundSimulator.dmg`
 2. Drag **Solar Sound System Simulator** into the **Applications** folder shortcut
 3. Eject the disk image
 4. Launch from Applications or Spotlight
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First-launch Gatekeeper warning

Because the app is not notarized with Apple, macOS may block it the first time.

To allow it:

- Right-click the app → **Open** → **Open** (one-time confirmation), or
 - System Settings → Privacy & Security → scroll down → **Open Anyway**
-

File layout after build

```
ForMac/  
├─ setup.py           # py2app configuration  
├─ build_dmg.sh       # build + package script  
├─ README_mac.md     # this file  
├─ AppIcon.icns       # (optional) place your icon here before building  
├─ build/             # intermediate py2app output (auto-created)  
├─ dist/  
│   └─ Solar Sound System Simulator.app  
└─ SolarSoundSimulator.dmg ← distribute this
```

Adding a custom icon (optional)

1. Create a 1024 × 1024 px PNG of your icon.
2. Convert it to `.icns` :

```
mkdir icon.iconset  
sips -z 512 512 icon.png --out icon.iconset/icon_512x512.png  
# (repeat for other sizes if desired)  
iconutil -c icns icon.iconset -o ForMac/AppIcon.icns
```

3. Re-run `./build_dmg.sh` — the icon will be embedded in the `.app`.
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Troubleshooting

Symptom	Fix
<code>import tkinter</code> fails	Use python.org Python or <code>brew install python-tk@3.12</code>
<code>py2app</code> not found	Run <code>pip install py2app</code> inside your active venv
App opens then immediately closes	Run from Terminal to see the error: <code>open -a "Solar Sound System Simulator" --args</code>
Gatekeeper blocks permanently	See Gatekeeper section above